DEFINITION

Banana Potential

A squared distance-to-segment potential with a quadratic vertical bend.

COMPACT FORM

$$\boxed{F_b(x,y) = \left(x - \Pi_{[0,1]}(x)\right)^2 + \left[y + b\left(x - \frac{1}{2}\right)^2\right]^2}$$

$$\Pi_{[0,1]}(x) = \min\{1, \max\{0, x\}\}\,, \qquad b \geq 0.$$

Equivalent piecewise form

Let $s_b(x,y) = y + b(x - \frac{1}{2})^2$. Then

$$F_b(x,y) = \begin{cases} x^2 + s_b(x,y)^2, & x < 0, \\ s_b(x,y)^2, & 0 \leq x \leq 1, \\ (x-1)^2 + s_b(x,y)^2, & x > 1. \end{cases}$$

DOMAIN

$$(x,y) \in [-1,2] \times [-2,2].$$

ZERO LEVEL SET

$$F_b(x,y) = 0 \quad \Longleftrightarrow \quad \begin{aligned} &0 \leq x \leq 1, \\ &y = -b\left(x - \tfrac{1}{2}\right)^2. \end{aligned}$$

The parameter b continuously controls the curvature: $b = 0$ gives a straight valley, while larger values produce a stronger banana shape.